ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY AND APPLIED SCIENCES

MID-SEMESTER EXAM

CCNA Security (CIT3372)

DATE: FRIDAY 27 OCTOBER 2023

TIME: 12:30 - 15:30 hours

INSTRUCTIONS:

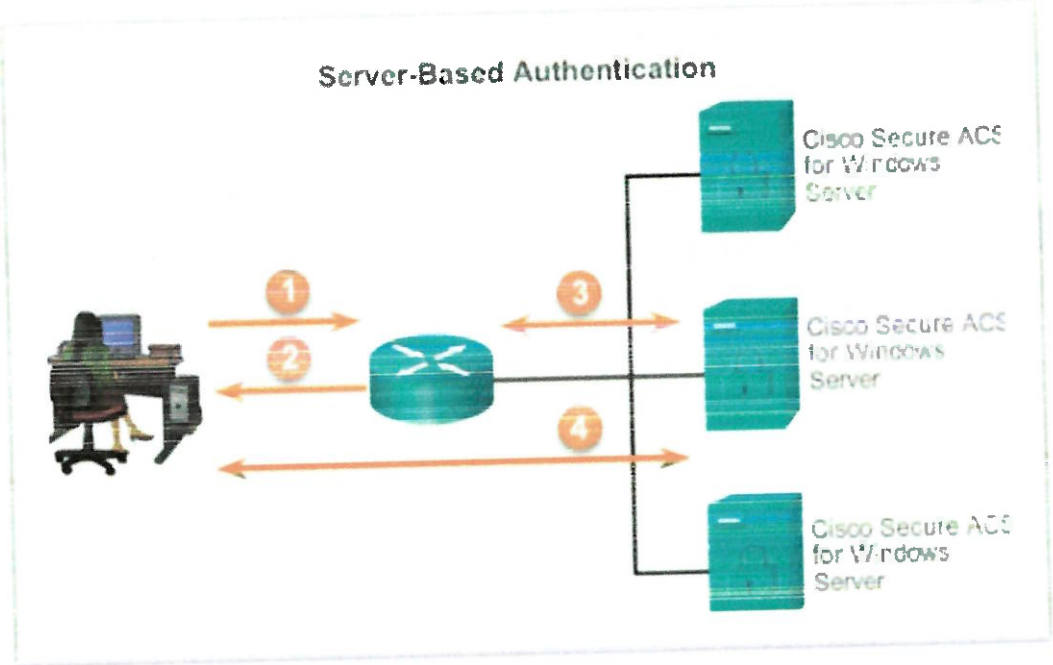
1. **Section A:** All questions in this section are compulsory and must be attempted.
2. **Section B:** Answer any **THREE** questions only and each question carries **20 Marks.**
3. This paper consists of **Five** (Questions) and carries a total of 100 marks.
4. Candidates must not turn this page until the invigilator tells them to do so.
5. Neat work with clear examples is high recommended.
6. Time allowed is 3 hours.

Section A: All questions in this section are compulsory and must be attempted.

QUESTION 1

Security implementation measures - A Case Study

Mr. Chizo Mukanda is employed by the Mungo Group of Companies, an organisation that boasts a workforce exceeding six thousand individuals. The individual in question has been employed within the Information and Communication Technology (ICT) department for a duration of five years. During this time, he has observed a recurring pattern wherein their systems are subjected to cyber-attacks on a quarterly basis. The decision was made to engage your services as a consultant possessing expertise in the field of cyber security. Your role will be to assist the company in identifying a sustainable resolution to the ongoing issues they have been encountering. It has been observed that users employ a local user account with plaintext passwords to connect onto their computers, lacking a robust authentication method for managing user access to the systems. The decision has been made to build an authentication server that will facilitate the authentication process for all personnel within the firm. The diagram presented below offers insights into the potential strategies for implementing security measures within a business.



a) Explain what is AAA and why is it important in network security? **[5 marks]**

b) List and explain the **THREE** main components of the AAA framework. **[6 Marks]**

c) Describe how AAA help in preventing unauthorized access to network resources? **6 Marks.**

d) Explain **FOUR** process of AAA authentication? **[8 Marks]**

e) List and explain **TWO** protocols used in CISCO for AAA authentication? **[5 Marks]**

f) State **THREE** ways on how AAA authorization work in controlling access to network resources. **[6 Marks]**

g) What is AAA accounting and why is it important? **[4 Marks]**

TOTAL MARKS 40

Section B: Answer any **THREE** questions only.

QUESTION 2

a) Mention **FIVE** modern network security threats that organizations face today? **[5 marks]**

b) Give **FOUR** points on how an organization can defend itself against ransomware attacks? **[4 Marks]**

c) Mention **FIVE** steps that organization can take to prevent phishing attacks? **[5 marks]**

d) List **SIX** measures the organizations can take to prevent data breaches? **[6 Marks]**

QUESTIONS 3

a) List and explain the **THREE** components of information security ensured by cryptography? **[6 Marks]**

b) Explain **FIVE** primary methods for mitigating malware? **[5 Marks]**

c) Describe **FIVE** ways on how the organizations can mitigate the impact of DDoS attacks? **[5 Marks]**

d) List **TWO** potential drawbacks or challenges of implementing AAA in a network? **[4 Marks]**

QUESTION 4

a) Explain the **THREE** benefits and limitations of the firewall. **[6 Marks]**

b) State the **FOUR** types of the firewalls. **[8 Marks]**

c) Give **TWO** reasons why ACL is important to be configure on the network. **[2 Marks]**

d) What are the differences between Standard and extended ACL. **[4 Mark]**

QUESTION 5

a) List and explain **FOUR** types of Hackers. **[8 Marks]**

b) Explain the difference between IDS and IPS in CCNA security. **[2 Marks]**

c) List **TWO** advantages and disadvantages of IDS and IPS. **[8 Marks]**

d) List the **Two** types of categories of cyber-attacks? **[2 Marks]**